

A Coherence-Gated Symmetron Model with Axial Spin Coupling Near Current Fifth-Force Bounds

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Abstract

We propose a theoretical extension of symmetron scalar field models in which the effective coupling to matter depends on macroscopic axial spin coherence. The model introduces a relativistic axial-vector coupling between a scalar field and fermionic spin currents, producing a continuous state-dependent interaction strength. The resulting force law takes the form of a Yukawa-type correction to Newtonian gravity. We discuss the theoretical structure, viable parameter window, experimental constraints, and signal-to-noise considerations. No experimental detection is claimed.

1 Introduction

Scalar field extensions to gravity have been studied as possible explanations for dark energy, screened fifth forces, and modifications to General Relativity. Symmetron models provide a mechanism in which scalar fields are screened in high-density environments and active in low-density environments.

This paper proposes a coherence-gated extension of the symmetron framework. The central idea is that the effective coupling may depend on a macroscopic axial spin order parameter. Ordinary random-spin matter produces negligible enhancement, while engineered coherent spin states may increase the effective interaction strength.

2 Theoretical Framework

We consider the action

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - V(\phi) + \frac{\phi^2}{2M^2} T^\mu_\mu + g_s \phi \bar{\psi} \gamma^\mu \gamma^5 \psi n_\mu \right]. \quad (1)$$

Here, ϕ is the scalar field, M is the symmetron coupling scale, T^μ_μ is the trace of the stress-energy tensor, and the final term represents an axial coupling to fermionic spin currents.

The symmetron potential is

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4 + V_0. \quad (2)$$

The effective potential is

$$V_{\text{eff}}(\phi) = \frac{1}{2} \left(\frac{\rho}{M^2} - \mu^2 \right) \phi^2 + \frac{1}{4}\lambda\phi^4 + V_0. \quad (3)$$

The critical density is

$$\rho_{\text{crit}} = \mu^2 M^2. \quad (4)$$

When $\rho < \rho_{\text{crit}}$, symmetry breaking occurs and the field obtains the vacuum expectation value

$$\phi_{\text{vac}} = \frac{\mu}{\sqrt{\lambda}} \sqrt{1 - \frac{\rho}{\rho_{\text{crit}}}}. \quad (5)$$

3 Coherence-Dependent Coupling

We define an axial order parameter

$$\mathcal{O} = \langle \bar{\psi} \gamma^\mu \gamma^5 \psi \rangle n_\mu. \quad (6)$$

The effective coupling is written as

$$\alpha_{\text{eff}} = \frac{\phi_{\text{vac}}^2}{M^2} \left(\frac{\mathcal{O}}{\mathcal{O}_{\text{max}}} \right). \quad (7)$$

For random spin states, $\mathcal{O} \approx 0$, giving negligible coherence enhancement. For engineered coherent spin states, $\mathcal{O}$ approaches $\mathcal{O}_{\text{max}}$ and the effective coupling increases.

4 Effective Force Law

The modified force law is

$$F(r) = -\frac{Gm_1m_2}{r^2} \left[1 - \alpha_{\text{eff}} \left(1 + \frac{r}{\lambda_\phi} \right) e^{-r/\lambda_\phi} \right]. \quad (8)$$

Here, λ_ϕ is the interaction range.

5 Parameter Space

A representative working point is

$$M \sim 10^9 \text{ GeV}, \quad \mu \sim 10^{-3} \text{ eV}, \quad \lambda \sim 10^{-6}, \quad \lambda_\phi \sim 5 \text{ cm}. \quad (9)$$

The baseline coupling is chosen to satisfy

$$\alpha_{\text{baseline}} \lesssim 2 \times 10^{-5}. \quad (10)$$

The coherence-enhanced regime may reach

$$\alpha_{\text{coherent}} \sim 5 \times 10^{-5} \text{ to } 10^{-4}. \quad (11)$$

6 Constraint Figure

7 Signal and Noise

For two test masses $m_1 = m_2 = 1 \text{ kg}$ separated by $r = 5 \text{ cm}$, the expected force scale is

$$F_{\text{signal}} \sim 10^{-13} \text{ to } 10^{-12} \text{ N}. \quad (12)$$

Schematic constraint plot.

The horizontal axis represents the interaction range λ_ϕ on a logarithmic scale. The vertical axis represents the effective coupling α_{eff} on a logarithmic scale.

The Eot-Wash torsion-balance experiments dominate constraints near centimeter scales. Around $\lambda_\phi \sim 5$ cm, the relevant bound is approximately $\alpha \lesssim (2 - 5) \times 10^{-5}$, depending on the analysis. MICROSCOPE strongly constrains longer-range equivalence-principle-violating interactions. Atom interferometry and Casimir-type experiments constrain shorter ranges.

The baseline model point is placed near $\lambda_\phi = 5$ cm and $\alpha_{\text{baseline}} \lesssim 2 \times 10^{-5}$. The coherence-enhanced regime may extend to $\alpha_{\text{coherent}} \sim 5 \times 10^{-5}$ to 10^{-4} under engineered axial spin coherence.

Figure 1: Schematic description of the model parameter space relative to current fifth-force constraints.

State-of-the-art force sensitivity may approach

$$F_{\text{noise}} \sim 10^{-14} \text{ N} \quad (13)$$

under highly controlled conditions.

Dominant systematic effects include magnetic gradients, thermal noise, seismic coupling, electrostatic patch forces, and residual gas effects. Magnetic systematics are especially important because the proposed coherence mechanism involves spin alignment and external driving fields.

8 Experimental Considerations

A possible experiment would require a low-noise torsion balance or levitated force sensor, strong magnetic shielding, controlled spin-coherence modulation, and reversal protocols. The key experimental signature would be a force correlated with the axial order parameter rather than with ordinary mass distribution alone.

A convincing detection would require distance scaling, coherence modulation, spin-orientation reversal, and suppression of magnetic and electrostatic backgrounds.

9 Discussion

The model is internally consistent as a phenomenological extension of symmetron scalar field theory. Its viable parameter space is narrow and lies near current experimental bounds. The baseline coupling must remain below existing fifth-force constraints, while the coherence-enhanced interaction must be absent in ordinary random-spin matter.

This work does not claim experimental confirmation. It proposes a testable theoretical framework for coherence-dependent scalar interactions.

10 Conclusion

We have presented a coherence-gated symmetron model with axial spin coupling. The model produces a Yukawa-type correction to Newtonian gravity whose effective strength depends on a macroscopic axial order parameter. The framework is speculative but testable and motivates further theoretical and experimental study.

References

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Disclaimer

This paper presents a theoretical model only. No experimental detection is claimed. The model is speculative and lies near current experimental constraints.